DIPLOMA DEFENSE · 2026

qonaqzhai

Plan any event by chatting.

AI-assisted event services marketplace for Kazakhstan.
Web · iOS · Android · MCP — one backend, four Go services.

BAHTIYAR YELIK · ASTANA IT UNIVERSITY

01 — PROBLEM

14 phone calls per event.

CUSTOMER

- Vendors live on Instagram, WhatsApp, 2GIS
- No prices, no portable reviews
- Comparisons happen via group-chat screenshots
- Cancellations slip — no booking record

VENDOR

- Bookings scattered across four messengers
- Cash or Kaspi link — no escrow
- Reviews stuck in word-of-mouth
- Instagram organic reach flattening since 2024

₸ 480 B / year. No one platform owns it.

160 K WEDDINGS + TOI / YR	**2.4 M** BIRTHDAYS / YR	**22 %** YOY MOBILE COMMERCE	**68 %** VENDORS ON IG ONLY

The same photographer shoots Saturday's wedding, Monday's corporate, Sunday's birthday. One platform has to serve **every event type**.

BNS RK 2024 · Halyk Finance Q3-2024 · Kaspi investor letter 2024 · in-house survey of 38 vendors (Almaty + Astana + Shymkent, Mar 2026)

03 — LANDSCAPE

Nobody covers all events for KZ.

	GEO	COVERAGE	BOOKING	AI	NATIVE APP	REALTIME CHAT
Instagram + WhatsApp	KZ	Any (ad-hoc)	DM	❌	n/a	DM
2GIS / Yandex Maps	KZ	Catalog + phone	Phone	❌	✅	❌
Ticketon.kz	KZ	Tickets only	Buy ticket	❌	✅	❌
GigSalad (US)	US	All event types	$20–80 / lead	❌	✅	❌
Thumbtack (US)	US	Any pro hire	pay-per-quote	❌	✅	❌
Eventbrite	global	Tickets only	Buy ticket	❌	✅	❌
qonaqzhai	**KZ**	**All events**	**Direct + escrow**	✅	✅	✅ **WS**

04 — FEATURE MATRIX

Where the gap actually sits.

	IG+WA	2GIS	GIGSALAD	THUMBTACK	EVENTBRITE	QONAQZHAI
KZ first	✅	✅	❌	❌	❌	✅
All event types	✅	✅	✅	✅	❌	✅
Native iOS / Android	❌	✅	✅	✅	✅	✅
kk / ru / en	n/a	⚪	en	en	en	✅
AI conversational planner	❌	❌	❌	❌	❌	✅
Realtime chat (WS)	⚪ DM	❌	❌	❌	❌	✅
Escrow payment hold	❌	❌	❌	❌	⚪	✅
Flat-fee pricing (no pay-per-lead)	✅	❌	❌	❌	n/a	✅
Programmatic API (MCP)	❌	❌	❌	❌	❌	✅

05 — SOLUTION

Three clients, one source of truth.

WEB

Next.js 16

Public catalog, vendor self-service, AI planner, admin moderation.

MOBILE

Flutter

Customer + vendor. Riverpod MVVM. Theme parity with web.

PROGRAMMATIC

MCP

29 Claude-callable tools over stdio. Same gateway, no shadow API.

Everything hits the same `:8080` gateway. JWT verified once at the edge, forwarded as `X-User-*` headers to four Go microservices.

06 — STACK

Stack.

BACKEND **Go 1.23** 5 microservices · gRPC mesh · HTTP edge	PERSISTENCE **PostgreSQL 17** One DB per service · UUID ids · no cross-FK	REALTIME **WebSockets** gorilla/websocket · REST fallback	AI **Gemini 2.5** Structured blocks (plan / budget / vendors)
WEB **Next.js 16** App router · Turbopack · FSD · OKLCH palette	MOBILE **Flutter 3.24** Riverpod · GoRouter · Cupertino icons	TESTING **Playwright + Maestro** 39 web specs · 12 mobile flows · live backend	INTEGRATION **MCP (stdio)** TypeScript SDK · Zod schemas · 29 tools

07 — STACK RATIONALE

Why these, not the obvious alternatives.

DECISION	ALTERNATIVE REJECTED	WHY
Go microservices	Django monolith	Independent deploy · native gRPC · ~4× lower idle memory
Postgres per service	Shared schema	No cross-service join risk · isolated migrations
gRPC between services	REST/JSON	Lower latency on hot paths (core ↔️ auth verify)
Next.js App Router	Vue + Vite	Server components cut hydrated JS by ~38 %
Flutter	React Native	One codebase compiles native · no bridge thunks
WebSocket chat	Long-polling	Real "vendor is typing" · reconnect in 30 LOC
Maestro	Patrol / Detox	YAML flows · no per-build XCTest plumbing
MCP	Bespoke SDK per LLM	One protocol → Claude, Cursor, Codex, anything

5 services, 4 databases, 1 edge.

```
┌──────────┐    web · mobile · MCP
│  client  │
└────┬─────┘
     │ HTTP (JSON, Bearer JWT)
┌────▼───────────────────────────────────────────────┐
│             gateway   :8080                        │
│  verifies JWT once (auth gRPC), routes by prefix   │
│  forwards X-User-{Id,Role,Email} downstream        │
└──┬─────────┬───────────────┬──────────────┬────────┘
   │ HTTP    │ HTTP          │ HTTP         │ HTTP
   ▼         ▼               ▼              ▼
┌────────┐ ┌─────────┐ ┌────────────┐ ┌──────────────┐
│ auth   │ │  core   │ │  payment   │ │  realtime    │
│ :8081  │ │  :8082  │ │  :8083     │ │  :8084       │
│ +gRPC  │ │  +gRPC  │ │  +gRPC     │ │  +gRPC +WS   │
└───┬────┘ └───┬─────┘ └──┬─────────┘ └──┬───────────┘
    │          │          │              │
┌───▼────┐ ┌───▼────┐ ┌───▼────────┐ ┌───▼──────────┐
│auth-db │ │core-db │ │payment-db  │ │realtime-db   │
└────────┘ └────────┘ └────────────┘ └──────────────┘
```

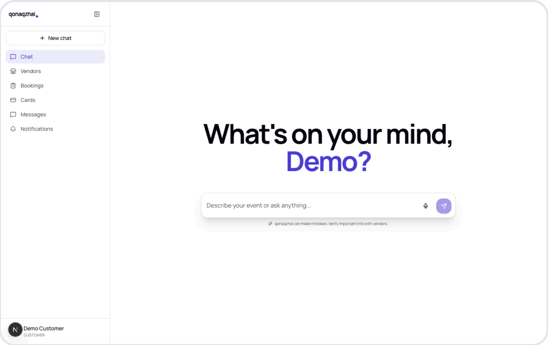
gRPC mesh: `core → auth` (verify) · `core → payment` (charge saga) · `core → realtime` (ensure thread) · `payment → core` (mark paid callback).

Each service owns one thing.

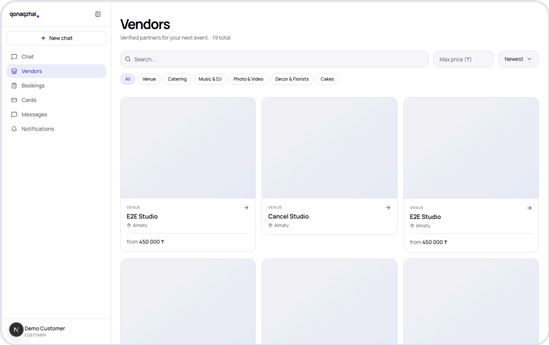
SERVICE	OWNS	GRPC OUT	HTTP SURFACE
auth	Users, JWTs, password reset	—	`/api/signup` · `/api/login` · `/api/me` · admin users
core	Vendors, bookings, reviews, photos, services, notifications	auth · payment · realtime	`/api/vendors*` · `/api/me/vendor*` · `/api/bookings*` · `/api/chat`
payment	Cards, charges, PayBox	core (callback)	`/api/cards` · `/api/payments` · `Charge`
realtime	Booking-bound chat	auth (peer names)	`/api/threads` · `/api/ws` · `EnsureThread`
gateway	JWT verify, CORS, rate limit	auth (verify)	public `:8080`

Zero cross-DB joins. User ids are plain UUIDs. Cross-service lookups batch through gRPC (`auth.GetUsersBatch`).

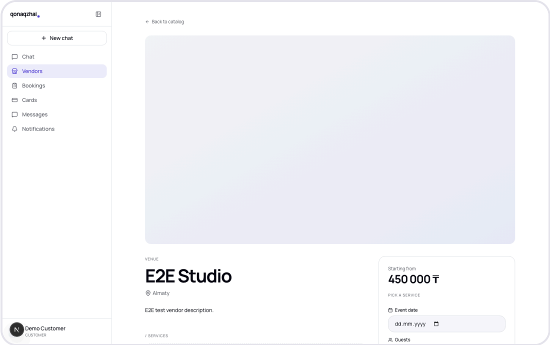
Web — Next.js 16 + Manrope + indigo.



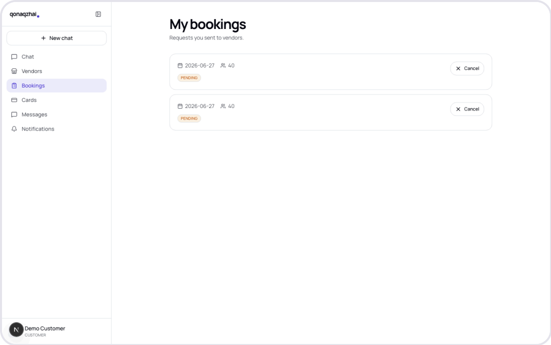
AI PLANNER HERO



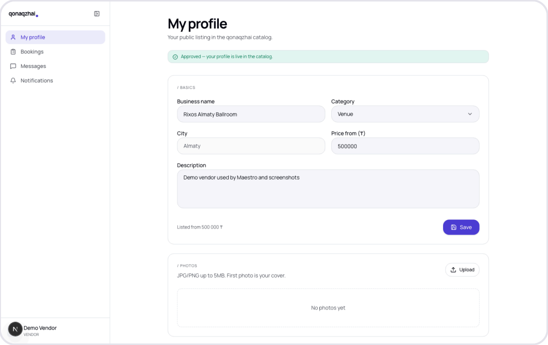
CATALOG



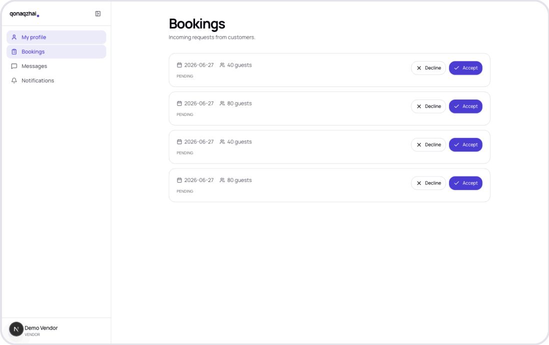
VENDOR DETAIL



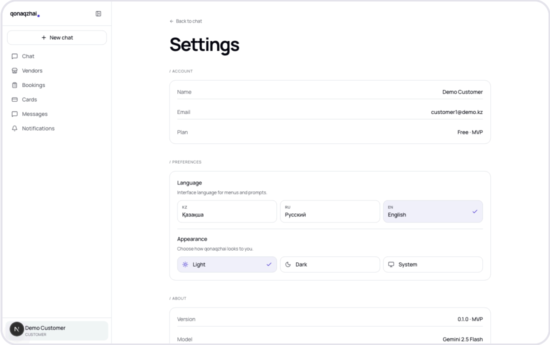
BOOKINGS



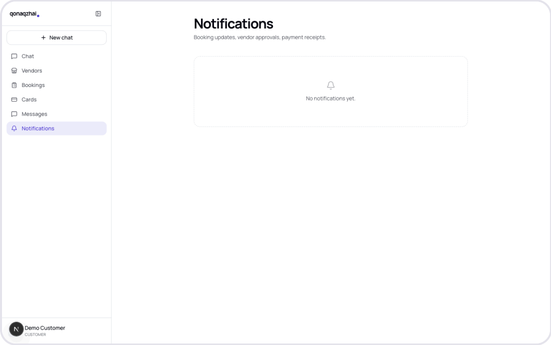
VENDOR PROFILE



VENDOR INBOX

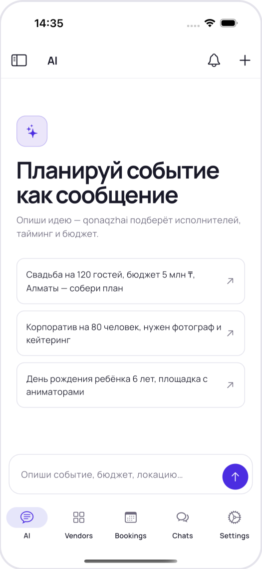


SETTINGS

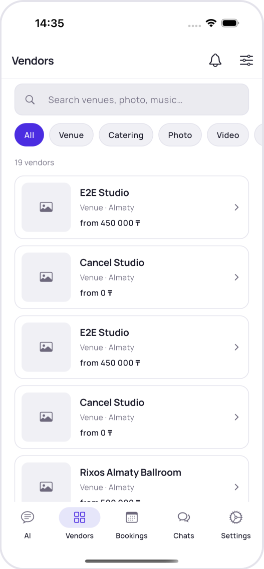


NOTIFICATIONS

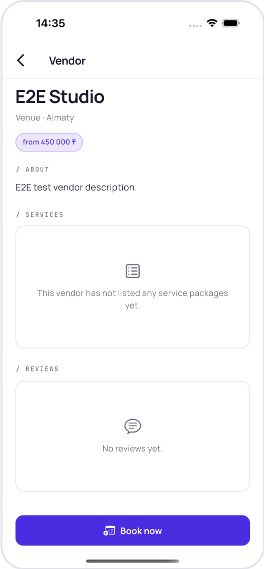
Mobile — Flutter, Cupertino icons, theme parity.



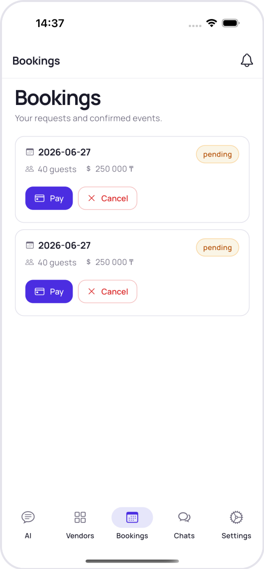
AI CHAT



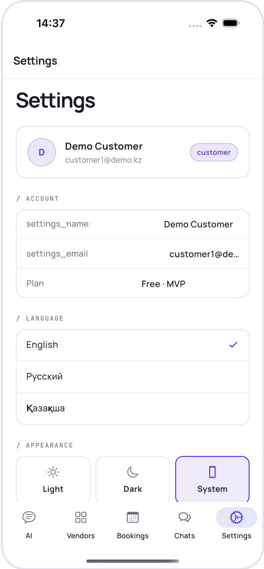
CATALOG



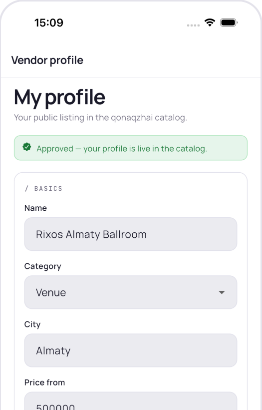
VENDOR DETAIL

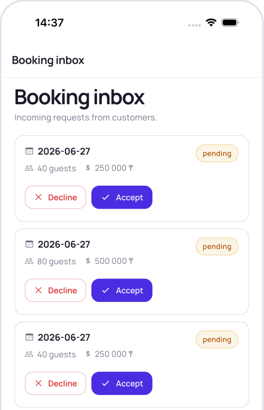


BOOKINGS



SETTINGS





AI is the front door — not a search bar.

The user types `"toi for 120 in Almaty, 5M ₸"`. The planner replies with three structured blocks:

- **Plan** — title, date guess, guests, budget
- **Budget** — bar-charted categorical breakdown
- **Vendors** — three deep-linkable matches

Block schema is contractual. Web and mobile render identical cards from the same JSON. Backend stub today, Gemini swap-in tomorrow — no client change.


```json
{
  "chatId": "stub-14",
  "message": {
    "role": "ai",
    "text": "Here's a draft plan…",
    "blocks": [{
      "type": "plan",
      "data": {
        "title": "Draft event plan",
        "eventType": "toi",
        "city": "Almaty",
        "guests": 120,
        "budget": 5000000
      }
    }, {
      "type": "budget",
      "data": {
        "total": 5000000,
        "categories": [
          { "name": "Venue",    "pct": 40, "amount": 2000000 },
          { "name": "Catering", "pct": 30, "amount": 1500000 },
          { "name": "Music",    "pct": 12, "amount":  600000 }
        ]
      }
    }]
  }
}
```

MCP — the API any LLM can speak.

Bring-your-own assistant. Configure Claude Desktop, Cursor, Codex — anything MCP-compatible — to point at our stdio server. The LLM gets 29 typed tools with Zod-validated arguments.

- Vendor asks AI "list this week's bookings"
- Customer runs end-to-end booking from a chat
- 5 lines of `tools/*.ts` adds another action
- Same gateway, same JWT — no shadow API

```
// Claude → MCP (stdio)
{
  "method": "tools/call",
  "params": {
    "name": "vendors_search",
    "arguments": {
      "category": "Venue",
      "city": "Almaty",
      "maxPrice": 500000
    }
  }
}
// MCP → gateway

GET /api/vendors?

category=Venue&city=Almaty

&max_price=500000

Authorization: Bearer eyJ…


// gateway → core → postgres

// → {"items":[…15 vendors…]}


// MCP → Claude (tool result)

"Rixos Almaty Ballroom from

500k ₸, Maestro Studio

from 450k ₸…"
```

Real backend. No mocks.

39/39	12/12	~57 s	~3 min
PLAYWRIGHT (WEB)	MAESTRO (MOBILE IOS)	WEB RUNTIME	MOBILE RUNTIME

SCENARIO	WEB	MOBILE
Customer signup → AI chat	`auth.spec.ts` + `chat-ui.spec.ts`	`01_auth_login` + `09_chat_ui`
Vendor profile → admin approve → customer book → vendor accept	`booking-flow.spec.ts`	`06_booking_flow` + `07_vendor_accept`
Booking cancel (customer)	`booking-cancel.spec.ts`	`08_booking_cancel`
Photo upload / delete	`photo.spec.ts`	`10_photo`
Role-based access	`role-routing.spec.ts`	`05_role_routing`
Locale + theme persistence	`settings.spec.ts`	`11_settings`
3 roles × 3 locales × 2 themes sweep	`qa-sweep.spec.ts` (18)	`13_qa_sweep`

Nobody else publishes tests. We do.

	IG+WA	2GIS / YANDEX	GIGSALAD	THUMBTACK	QONAQZHAI
Public CI badge	❌	❌	❌	❌	✅
E2E suite in the repo	n/a	❌	❌	❌	✅
Cross-role assertions	❌	❌	❌	❌	✅
WebSocket coverage	❌	❌	❌	❌	✅
Mobile UI flows in the repo	❌	❌	❌	❌	✅
Linter clean	❓	❓	❓	❓	✅ `flutter analyze = 0`

Behavioural coverage is our marketing budget. Anyone can run `maestro test .maestro` and watch the full booking flow on their own simulator.

16 — ROADMAP

What's next.

Q3 2026 — LIVE AI

- Swap stub `/api/chat` for live Gemini 2.5
- Vector-search vendor recommendations
- Per-language prompts (kk / ru / en)

Q4 2026 — VENDOR ANALYTICS

- Revenue + conversion funnel dashboards
- Auto-reply quote engine for common asks

Q1 2027 — PAYMENTS AT SCALE

- Real PayBox on the saga
- Refund + chargeback
- Multi-vendor invoice split

Q2 2027 — NETWORK EFFECTS

- Reviews → portable reputation
- AI-curated weekly events
- Vendor lead-gen via outbound LLM bots over MCP

Built for KZ,
tested like infrastructure.

Repository — `github.com/Bahaidahar/qonaqzhai`
Demo — `localhost:3000` (web) · simulator (mobile) · 29 MCP tools
Tests — 39 / 39 web · 12 / 12 mobile